GOODS.NG

Buy. Sell. Trade with Confidence.

Product, Architecture & Functional Requirements

A single go-live documentation set for the Goods.ng marketplace: Product Requirements Document (PRD), Technical Architecture Document (TAD), and Functional Requirements Document (FRD).

VERSION
1.0

DATE
8 November 2026

STATUS
Go-live planning baseline

MARKET
Nigeria · NGN

Confidential — internal product & engineering document

DOCUMENT CONTROL

About this document

This document consolidates three deliverables that are normally maintained separately. They are bound together because the launch decision depends on all three: what the product must do (PRD), how the system is built and operated (TAD), and the testable behaviour expected of each feature (FRD). The appendix contains the integration go-live checklist.

Product	Goods.ng — multi-vendor marketplace for Nigeria
Owner	Product & Engineering, Goods.ng
Audience	Founders, product, engineering, QA, security, operations, compliance, and integration partners
Source of truth	Repository `docs/`, application code, migrations and automated tests as of 8 November 2026
Verification baseline	205 backend tests, 5 frontend unit tests, 6 Playwright E2E scenarios, ruff/eslint/tsc clean, production build passing
Change control	Any change to scope, architecture or an external provider requires a version bump and a launch-gate review

Status vocabulary used throughout

Badge	Meaning
LIVE	Implemented in code and covered by automated tests; runs against mock or local infrastructure in development.
CONFIG	Implemented, but production requires credentials, a commercial account, DNS, or a provider dashboard step before it is live.
MOCK	Only a mock adapter exists. A licensed partner adapter and contract are required before live use.
NOT BUILT	No implementation yet; required or recommended for launch.
OPTIONAL	Not required for first launch; planned for growth or scale.

> **Important launch caveat**
> Goods.ng does not hold customer funds. Protected Checkout and Goods Flex currently run against **mock escrow and mock lending partners**. Production boot explicitly refuses mock financial providers. The product must not be described as holding funds or lending until licensed partner adapters are signed, implemented and certified.

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PART I

Product Requirements Document

What Goods.ng is, who it serves, what is in the first release, how it makes money, and the measurable gates that must be met before launch.

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1. Purpose and scope

This PRD defines the first production release of Goods.ng, a Nigeria-first, multi-vendor marketplace that combines classifieds-style discovery with in-app chat, price negotiation, identity and business verification, escrow-protected checkout, and a buy-now-pay-later product called **Goods Flex**. It is written for the team that will take the product from a working development build to a licensed, operated, revenue-generating service.

The PRD covers product intent, users, scope, economics, journeys, priorities and launch gates. Implementation detail belongs in the TAD; testable feature behaviour belongs in the FRD. Where the three documents disagree, the FRD is the acceptance authority, the TAD is the implementation authority, and the PRD is the priority authority.

2. Executive summary

Nigerian online commerce is split between classifieds platforms where buyers and sellers transact directly — with high fraud, no payment protection and no accountability — and traditional e-commerce sites that control inventory but offer limited negotiation, weak local-seller tooling and little flexibility on payment. Goods.ng occupies the middle: a trusted marketplace where every listing can be discussed, negotiated and paid for through a protected flow, while sellers keep their independence and build a verifiable reputation.

The product is already broad and functionally deep: 20 backend modules, 185 REST endpoints, 54 database tables, 17 root categories (91 total), protected checkout with a double-entry ledger, negotiation, messaging, seller verification, reviews, subscriptions, per-listing promotions, advertising feeds, and a 20-area admin console. The launch gap is not feature breadth; it is the **external integration layer** (licensed escrow and lending partners, payment certification, KYC, SMS/email, observability, payouts, logistics) and the legal/operational readiness that goes with it.

205
BACKEND TESTS PASSING

185
REST ENDPOINTS

54
DATABASE TABLES

17 / 91
ROOT / TOTAL CATEGORIES

3. Vision, mission and goals

Vision. Make everyday trade in Nigeria safe enough to do online without fear.

Mission. Give buyers and sellers the tools to discover, agree and complete a transaction with confidence: real listings, real people, protected money and a reputation that follows both sides.

Product goals for the first release

1. **Trust:** a buyer can pay through Goods.ng without losing money if the item is not delivered as described.
2. **Liquidity:** enough quality inventory and seller activity in the launch categories for a buyer to find something worth buying.
3. **Conversion:** reduce the friction between "I am interested" and "paid" through chat, offers, a protected checkout and instalment payments.
4. **Repeatability:** sellers return because the tools (storefront, analytics, promotions, verification) produce measurable sales.
5. **Operability:** the business can be run safely by a small team with audit trails, reconciliation and clear escalation paths.

4. Market context and problem

Problem	Current reality	Goods.ng response
Fraud and non-delivery	Buyers send bank transfers to strangers; sellers ship to buyers who dispute after delivery. Evidence is weak and recovery is rare.	Protected Checkout with a licensed escrow partner, server-side payment verification, delivery evidence, auto-release and a dispute workflow.

Problem	Current reality	Goods.ng response
Price negotiation is informal	Haggling happens on WhatsApp with no record; agreed prices are disputed and stale screenshots are unreliable.	Structured offers and counters with expiry, turn-taking, a 10-round cap and a checkout locked to the accepted offer.
Sellers are anonymous	Anyone can post a listing; reputation is a profile picture and a phone number.	Phone, identity and business verification tiers; verified badges; reviews only from delivered orders.
Payment inflexibility	Buyers must pay in full; sellers wait for settlement; no formal instalment option exists for everyday goods.	Goods Flex: credit assessment from open-banking data, a mandatory guarantor, transparent instalments, and one running plan per customer.
Discovery is noisy	Feeds are dominated by spam and repeat posts; filters are shallow and search misses synonyms.	PostgreSQL full-text search with weighted fields and prefix matching, category facets, premium ranking, and quality-gated sponsored placements.
Seller growth is manual	No analytics, no promotion products, no way to buy visibility for a single listing.	Seller dashboard, per-listing TOP and Boost Premium packages, subscriptions with featured credits, and advertising feeds.

5. Personas

Persona	Needs	Success looks like	Key surfaces
Adaeze — buyer Lagos, 29, salaried	Find a genuine phone or appliance at a fair price, ask questions, negotiate, pay safely, and know when it will arrive.	Buys within one session; item matches the listing; money is released only after delivery.	Home, search, item, chat, offers, checkout, orders
Chinedu — business seller Ikeja, phone shop	List stock quickly, look credible, answer buyers, close sales, promote individual products without a full subscription.	Verified badge, steady enquiries, measurable views, repeat buyers, paid promotions that pay back.	Seller dashboard, products, orders, analytics, promotions, storefront
Aisha — fashion seller Abuja, made-to-order	Sell unique pieces, manage custom orders, build a brand, use WhatsApp for relationship selling.	Storefront visits convert to chats; premium contact controls drive direct sales.	Storefront, listings, chat, subscription
Ngozi — premium seller Port Harcourt, appliances	Priority visibility, analytics, featured credits and a trustworthy checkout for high-ticket items.	Premium ranking, more qualified leads, lower dispute rate.	Premium, analytics, ads, promotions
Tunde — auto dealer Lekki, vehicles	High-value listings, inspection-driven trust, serious buyers only.	Qualified enquiries; protected checkout for large transactions; clear delivery/collection terms.	Listing editor, chat, orders, verification
Guarantor Friend or family of a Flex buyer	Understand exactly what they are guaranteeing, for how long, and how to withdraw.	Informed consent, a capped amount, SMS notification and a clear record.	Guarantor consent link
Operations admin Support, trust & safety	Resolve disputes, moderate listings, verify sellers, reconcile money, investigate abuse.	Every action is permissioned, audited and reversible where appropriate.	Admin console (20 areas)

6. Product scope

In scope — first release

- Mobile-first public marketplace with SSR/SEO.
- 17 root categories, 91 categories, 36 states + FCT.
- Seller onboarding, verification tiers, storefronts.
- Listing creation, image pipeline, moderation, stock.
- Search, filters, facets, favourites, recommendations.
- Product-linked chat and structured negotiation.
- Protected Checkout, escrow state machine, ledger.
- Orders, delivery updates, evidence, auto-release.
- Disputes, refunds and four-eyes approval.
- Goods Flex with guarantor and open-banking assessment.
- Reviews from delivered orders only.
- In-app notifications + email/SMS outbox.
- Subscriptions and per-listing promotions.
- Google/Meta product feeds and tracking.
- Admin console with RBAC and audit logs.

Out of scope — first release

- Native iOS/Android apps (responsive web and PWA-ready only).
- Goods.ng-owned inventory or first-party warehousing.
- In-house wallet, stored value or direct fund custody.
- International shipping and multi-currency checkout.
- Live video shopping or livestream auctions.
- Full logistics network; manual tracking initially.
- Automated seller payouts (partner-dependent).
- Partial refunds in the UI (model supports amounts).
- Dedicated search engine (Postgres FTS is sufficient).
- WebSocket messaging (5-second polling initially).
- Fraud-scoring vendor and address verification.
- Business accounting or invoicing suite.

7. Value proposition

For	Value	Proof point in the product
Buyers	Shop strangers safely, negotiate transparently, pay in a way that protects the money, and get recourse.	Protected Checkout, verified sellers, structured offers, delivery evidence, disputes, reviews.
Sellers	A credible storefront, qualified buyers, tools to grow, and a path to higher visibility without giving up independence.	Verification badges, analytics, premium placement, per-listing promotions, storefront and chat.
Flex customers	Afford everyday purchases through instalments with clear terms and a human guarantor model.	Six-month bank assessment, transparent schedule, mandatory guarantor consent, one-plan rule.
Partners	A regulated-ready marketplace that routes protected transactions to licensed escrow and lending partners with clean reconciliation.	Provider interfaces, append-only ledger, webhook verification, reconciliation reports.

8. Business model

Revenue stream	Mechanism	Default	Status
Marketplace commission	Percentage of the item subtotal on a completed protected order, deducted before seller settlement.	5.0%	LIVE mock settlement
Buyer protection fee	Percentage of subtotal, capped per order, shown separately at checkout.	1.5% · cap ₦5,000	LIVE mock settlement
Premium subscriptions	Monthly, quarterly and annual seller plans with featured credits and premium tools.	₦15k / ₦40k / ₦150k	LIVE mock payments
Per-listing promotions	One-off TOP 7-day, TOP 30-day, or Boost Premium 28-day visibility for a single listing.	₦3,099 / ₦9,999 / ₦31,999	LIVE mock payments

Revenue stream	Mechanism	Default	Status
Advertising	Promoted placements and catalog ads on Google/Meta; tracking and product feeds.	Campaign-defined	CONFIG

> **Pricing governance**
> Commission, protection fee and cap are environment defaults and can be overridden in Platform Settings. Promotion package prices are server-side and overridable through the `promotion_packages` setting. Client-supplied prices are always ignored.

9. Key user journeys

9.1 Buyer: discover → agree → pay → receive

1. Lands on home or a category; searches or browses; filters by state, price, condition and brand.
2. Opens an item page with photos, seller verification, delivery options and protection terms.
3. Starts a product-linked chat or makes an offer. Counter-offers are structured and time-boxed.
4. Accepts the price (or the seller accepts the offer); checkout is locked to that accepted offer for 24 hours.
5. Chooses delivery, sees the subtotal, delivery fee, protection fee and total; pays through the licensed provider.
6. Goods.ng verifies the payment server-to-server; the escrow partner holds the funds; the seller is notified.
7. Seller accepts, processes and dispatches with evidence; buyer confirms delivery or the order auto-releases after the configured window.
8. Buyer leaves a review; seller is paid through the partner; the ledger records commission and protection fee.

9.2 Seller: onboard → list → sell

1. Creates an account, becomes a seller, adds store name, location and contact details.
2. Completes phone, identity or business verification; uploads supporting documents privately.
3. Creates a listing with images, condition, price, stock, delivery options and BNPL eligibility; auto-title suggests a keyword-rich title.
4. Receives enquiries and offers; responds in chat; accepts or declines orders.
5. Tracks views, favourites, enquiries and earnings; optionally buys a promotion or a subscription.

9.3 Goods Flex: assess → guarantee → apply → repay

1. Customer verifies phone and BVN, connects a bank account read-only, and receives an affordability assessment.
2. Names a guarantor, who receives an SMS link, sees the amount and terms, and consents explicitly.
3. Applies for a plan; the licensed lender returns a decision and terms; the customer accepts.
4. Pays the down payment; the partner disburses; the order follows the same protected checkout path.
5. Repays instalments; reminders and overdue states are handled by the worker; only one plan can run at a time.

9.4 Admin: trust and money operations

1. Moderates listings and reviews; verifies sellers; suspends abusive accounts.
2. Resolves disputes with evidence, issues refunds under four-eyes approval, and reconciles payments, escrow and ledger.
3. Grants subscriptions, adjusts platform settings, manages staff roles, and reviews the append-only activity log.

10. Functional epics and priorities

Epic	Outcome	Priority	Launch state
Authentication & account	Secure sign-up, login, OTP, password reset, 2FA, sessions.	P0	LIVE
Catalog & listings	Categories, listing lifecycle, images, stock, moderation.	P0	LIVE
Search & discovery	Full-text search, facets, ranking, favourites, recommendations.	P0	LIVE
Messaging	Product-linked conversations, unread counts, reports.	P0	LIVE
Negotiation	Offers, counters, expiry, accepted-price checkout.	P0	LIVE

Epic	Outcome	Priority	Launch state
Payments	Provider abstraction, verification, webhooks, refunds.	P0	CONFIG
Protected checkout & escrow	Funds held by a licensed partner; release/refund state machine.	P0	MOCK
Orders & delivery	Seller acceptance, dispatch, evidence, auto-release.	P0	LIVE
Disputes & refunds	Evidence, admin resolution, four-eyes refund approval.	P0	LIVE
Seller verification	Phone, identity, business tiers; private document review.	P0	CONFIG
Goods Flex	Credit assessment, guarantor, lender decision, instalments.	P1	MOCK
Reviews	Verified-purchase reviews, replies, moderation.	P1	LIVE
Notifications	In-app inbox, preferences, email/SMS outbox.	P1	CONFIG
Subscriptions	Premium plans, featured credits, admin grants.	P1	LIVE
Promotions	Per-listing TOP and Boost Premium packages.	P1	LIVE
Ads & feeds	Google/Meta catalog feeds, pixel events, CAPI.	P2	CONFIG
Admin & staff	20 areas, granular RBAC, audit, reconciliation.	P0	LIVE
Trust & safety	Moderation, reports, direct-contact warning, suspension.	P0	LIVE

11. Success metrics and launch gates

North-star and supporting metrics

Metric	Definition	First-90-day target
Protected GMV	Value of orders paid through Protected Checkout, excluding refunds.	Baseline set in staging; grow week-on-week
Order completion rate	Completed orders ÷ paid orders.	≥ 90%
Dispute rate	Orders with a dispute ÷ paid orders.	< 3%
Time to first response	Median seller response to a new enquiry.	< 30 minutes in business hours
Listing quality	Share of active listings with 3+ photos and a 60+ character description.	≥ 80%
Verified seller share	Active sellers at identity or business tier.	≥ 60%
Flex repayment health	Instalments paid on time ÷ due instalments.	≥ 95%
Search success	Sessions with a search and an item view ÷ search sessions.	≥ 45%
p95 API latency	Read endpoints at target concurrency.	< 500 ms
Error rate	5xx responses ÷ total API responses.	< 0.5%

Launch gates (all must be true)

#	Gate	Owner
G1	Licensed escrow/payment partner contract signed; adapter implemented and certified in sandbox; production boot passes.	Finance / Eng
G2	Licensed lender contract signed; FinancingProvider adapter implemented; FCCPC and legal review complete.	Risk / Legal
G3	Primary payment provider (Paystack or Flutterwave) certified in sandbox and live; webhooks, settlement and reconciliation verified.	Finance / Eng
G4	KYC provider live for BVN/NIN; consent and data-retention copy approved.	Compliance

#	Gate	Owner
G5	SMS and email providers live; sender ID/domain authenticated; OTP and transactional delivery proven.	Eng
G6	Managed Postgres with PITR, managed Redis, S3 + CDN, and at least two API instances deployed.	Platform
G7	Sentry/error tracking, structured logs, uptime checks and alerts live; on-call rota agreed.	Platform
G8	External penetration test complete; critical/high findings closed.	Security
G9	Legal review of Terms, Privacy, Refund & Dispute, Goods Flex and guarantor consent complete.	Legal
G10	Staging end-to-end: sign-up → listing → offer → payment → escrow → delivery → release → review, with real provider sandboxes.	QA

12. Assumptions, dependencies and risks

Assumptions

- Goods.ng will operate as a marketplace and agent, not as a bank, lender or escrow provider.
- A licensed partner is available and commercially viable for escrow and for BNPL.
- Buyers and sellers have smartphones and intermittent connectivity; the web experience must stay lightweight.
- NGN is the only currency and Nigeria the only market at launch.
- Sellers will accept a 5% commission and a buyer-paid protection fee in exchange for trust and demand.

Dependencies

- Escrow/lending partner contracts, sandbox access and legal sign-off.
- Payment provider approval and settlement account.
- KYC and open-banking provider approval, including NDPA-compliant consent.
- Termii sender ID approval and email domain authentication.
- Cloud accounts, DNS and SSL for goods.ng and admin.goods.ng.

Top risks

Risk	Impact	Mitigation
Escrow partner not signed in time	Protected Checkout cannot launch; core differentiator unavailable.	Start partner diligence immediately; keep a fallback licensed payment partner; do not launch unprotected checkout under the "Protected" brand.
BNPL regulatory friction	Goods Flex cannot launch or must be re-scoped.	Launch Flex behind a feature flag after FCCPC review; keep the product usable without it.
Fraud at onboarding	Chargebacks, fake listings, reputational damage.	Verification tiers, KYC, moderation, device/rate limits, delivery evidence, manual review for high-value orders.
Seller liquidity	Buyers find nothing to buy.	Seed launch categories with verified sellers; run supply acquisition before paid marketing.
Payment webhook outages	Orders stuck unpaid or unfulfilled.	Verify-before-trust, buyer return-page re-verification, expiry job that re-checks the provider, reconciliation alerts.
Operational overload	Disputes and KYC queues grow faster than the team.	Admin SLAs, canned responses, audit trails, four-eyes refunds, staffing plan for launch.
Data breach	Regulatory and reputational harm.	Encryption at rest, private KYC storage, least privilege, pen test, incident response, NDPA records.

13. Roadmap

Phase	Scope	Exit criteria
Now → launch	Integration layer, legal/compliance, observability, deployment, pen test, staging E2E.	All ten launch gates met.

Phase	Scope	Exit criteria
Launch + 30 days	Operate, monitor, fix funnel friction, tune ranking and fees, grow verified supply.	Stable error rate, dispute rate < 3%, completion ≥ 90%.
Launch + 90 days	Payout automation, logistics partners, partial refunds, push notifications, saved searches.	Seller payout cycle < T+1 after release; at least one carrier integrated.
Scale	Dedicated search, WebSockets/SSE, fraud scoring, mobile apps, more cities/categories.	Search p95 stable at 10× traffic; messaging no longer polling-bound.

14. Compliance and legal

- **NDPA 2023:** lawful basis, privacy notice, cookie consent, data-subject requests, retention schedule, DPO appointment, processor agreements.
- **FCCPC digital lending:** transparent charges, affordability, responsible collections, complaint handling, no dark patterns.
- **CBN-licensed partner:** funds held by a licensed escrow/payment partner; Goods.ng is not a deposit-taking institution.
- **Guarantor law:** enforceability, disclosures, cooling-off and withdrawal rights reviewed by counsel.
- **Consumer protection:** clear refund and dispute policy, seller obligations, prohibited items, takedown process.
- **Advertising:** claims must match the licensed reality; no "insured", "guaranteed" or "bank" language without partner approval.

PART II

Technical Architecture Document

How the system is structured, deployed, secured and operated, and the exact state of every external integration required for go-live.

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2. System context
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5. Backend architecture
6. Data architecture
7. API architecture
8. Identity and access
9. Payments and escrow
10. Goods Flex
11. Search
12. Messaging and negotiation
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14. Media pipeline
15. Background processing
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17. Security
18. Observability
19. Integrations and go-live
20. Scalability
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1. Architecture principles

1. **Modular monolith first.** One deployable API with clear module boundaries and a single database. Scale the process horizontally before splitting services.
2. **Provider interfaces at every regulated or external boundary.** Payments, escrow, lending, open banking, identity, storage, SMS and email are behind Protocols so a partner can be replaced without touching domain logic.

3. **Verify before trust.** A payment is real only after a server-to-server verification with an exact amount and currency match. Webhooks trigger verification; redirects and screenshots never do.
4. **The database is the source of truth.** Financial invariants are enforced by constraints, row locks, unique keys and triggers, not only by application code.
5. **Idempotency everywhere.** Checkout, webhooks, settlement, refunds, promotions and BNPL payments can be retried safely.
6. **Same-origin API.** The browser talks only to the Next.js origin, which proxies `/api/v1` and `/media` to the backend. This keeps the refresh cookie first-party and removes CORS from the production path.
7. **Operate what you build.** Structured logs, request IDs, health checks, reconciliation and audit trails are part of the feature, not an afterthought.

2. System context

```
Browser (mobile-first, SEO public pages)
  │  same-origin /api/v1/* and /media/*
  ▼
Next.js 16 (Vercel)  ── rewrite ──▶  FastAPI (stateless, N pods)
  │  SSR / PPR public pages            /api/v1 REST · JWT + RBAC
  │  client dashboards                 modules/* domain services
  ▼                                        │            │
PostgreSQL 15+ (source of truth) ◀─────────┘            │
Redis (cache, rate limits, single-flight) ◀────────────┘
  ▲
Worker (python -m app.worker): outbox, expiry, auto-release, reminders, feeds, CAPI

External, behind interfaces:
  PaymentProvider (mock | Paystack | Flutterwave)
  EscrowProvider (mock only today)
  FinancingProvider (mock only today)
  BankDataProvider (mock | Mono, untested)
  IdentityProvider (mock only today)
  Object storage (local | S3-compatible)
  SMS (console | Termii) · Email (console | SMTP | Postmark)
  Google OAuth · reCAPTCHA/Turnstile · Meta CAPI
```

3. Technology stack

Layer	Technology	Notes
Frontend	Next.js 16.4 (App Router, Cache Components/PPR), React 19.3, TypeScript 5, Tailwind 4	System fonts, streaming SSR, client dashboards on a central API client.
Backend	Python 3.12, FastAPI, Pydantic 2, SQLAlchemy 2, Alembic	185 REST endpoints, 20 modules, OpenAPI only in dev/test.
Database	PostgreSQL 15+	54 tables, full-text search, partial unique indexes, append-only triggers, deferred balance check.
Cache / limits	Redis 5+	Required in production. In-memory fallback in development. Rate limits and single-flight locks.
Worker	Same image, `python -m app.worker`	Postgres advisory locks make multiple replicas safe.
Storage	Local (dev) or S3-compatible + CDN	Pillow decode, WebP re-encode, watermark, private KYC prefix.
Tests	pytest, Vitest, Playwright, ruff, eslint, tsc	205 backend, 5 unit, 6 E2E; CI runs migrations round-trip and build.
Deploy	Vercel (frontend) + container platform (API/worker) + managed Postgres/Redis	Dockerfiles and Compose exist; not yet deployed.

4. Deployment topology

Component	Production shape	Scaling / resilience
Frontend	Vercel, root `frontend/`, `pnpm build`; rewrites `/api/v1` and `/media` to the API.	CDN edge, static/PPR pages, per-request streaming for data.
API	Container image from `backend/Dockerfile`, ≥ 2 instances behind HTTPS.	Stateless; horizontal scale. Pool: (10 + 20) × workers × replicas.
Worker	Same image, separate command, ≥ 1 replica.	Advisory locks prevent duplicate job execution; scale horizontally.
Database	Managed Postgres 15+, TLS, PITR, private network; PgBouncer (transaction pooling) as replicas grow.	Read replicas for browse/search before write saturation; 15 s statement timeout.
Redis	Managed Redis, private network.	Cache, rate limits, single-flight. Boot fails without it in production.
Storage	S3-compatible bucket, versioning, CDN for public media, no public access for `private/`.	Signed URLs for KYC; lifecycle rules; cross-region replication for private data.
DNS	goods.ng, admin.goods.ng; SSL managed by the platform.	Admin host carries `X-Robots-Tag: noindex`; optional edge IP allow-list.

> **Connection budget**
> Each API process opens up to `DB_POOL_SIZE + DB_MAX_OVERFLOW` connections. Total = that × `WEB_CONCURRENCY` × replicas. Size this against `max_connections`, or put PgBouncer in front. This is the first thing to tune when adding pods.

5. Backend architecture

The API is a modular monolith. Each module owns its router and business logic; cross-module coupling goes through service functions and the payment purpose registry. The payments module never imports orders, subscriptions or BNPL directly.

Layer	Responsibility	Examples
`core/`	Config, DB session, security primitives, errors, JSON logging with request IDs, cache/rate limiter, auth/RBAC dependencies.	Argon2, JWT, Fernet, Redis, trusted proxy IP resolution.
`models/`	SQLAlchemy mappings and constraints.	Users, sellers, catalog, messaging, commerce, ledger, misc.
`integrations/`	External provider Protocols and adapters.	Payments, escrow, BNPL, bank data, identity, storage.
`modules/`	Domain services, routers, state machines, policies.	auth, users, sellers, categories, products, subscriptions, messaging, negotiation, payments, orders, promotions, bnpl, reviews, notifications, admin, staff, stats, recommendations, ads, marketing.
`worker.py`	Scheduled jobs with advisory locks.	Outbox, expiry, auto-release, subscription/promotion expiry, BNPL reminders, feed/CAPI, purge.

Payment purpose registry

A single payment can satisfy different business purposes: an order, a subscription, a listing promotion, or a BNPL instalment. The payments module activates the correct handler on verified success. This keeps payment verification independent of the domain that initiated it.

Purpose	Effect on verified success	Ledger posting
Order / Protected Checkout	Confirm payment, fund escrow, reserve stock already held, move order to funds safeguarded.	Debit provider receipts / credit escrow obligations (via partner mirror).
Subscription	Activate plan, grant featured credits, set period.	Debit provider receipts / credit subscription revenue.
Promotion	Activate listing promotion, set tier/featured window.	Debit provider receipts / credit promotion revenue.
BNPL instalment	Mark instalment paid, update plan status, notify.	Recorded against the financing partner's position.

6. Data architecture

54 tables in six domains. Money is `Decimal` with at most two decimal places and is serialised as a string; it never touches floats. The database enforces the invariants that matter most.

Domain	Core tables	Key guarantees
Identity & access	users, profiles, refresh_tokens, otp_codes, two_factor, admin roles/permissions, staff invitations, audit logs	Token rotation and reuse detection; append-only audit; granular RBAC.
Seller & catalog	seller_profiles, verifications, categories, products, product_images, favourites, reports	One seller profile per user; slug uniqueness; image ownership; verification tiers.
Messaging & negotiation	conversations, messages, read receipts, negotiations, offers	One live negotiation per buyer/product (partial unique index); stale-offer rejection.
Commerce	orders, order_items, payments, escrow_records, escrow_events, refunds, disputes, disputes_messages, reviews	Total = subtotal + delivery + protection; one active refund; append-only escrow events.
Growth	subscription_plans, subscriptions, promotions, platform_settings, ads/events, interest signals	Premium derived from an active period; promotion tier mirrors an active paid promotion.
Flex & ledger	credit_profiles, bank_connections, credit_assessments, guarantors, applications, agreements, instalments, ledger_accounts, ledger_transactions, ledger_entries	One running plan; guarantor cap; immutable assessment snapshots; balanced ledger at commit.

Ledger design

The ledger is double-entry and append-only. A deferred constraint trigger rejects any unbalanced transaction at commit. Database triggers block updates and deletes on ledger, escrow-event and audit tables.

Event	Debit	Credit
Escrow funded	`asset:escrow_partner_holding`	`liability:escrow_obligations`
Released to seller	`liability:escrow_obligations`	`liability:seller_payable:<seller>`, `revenue:platform_commission`, `revenue:protection_fees`
Refunded	`liability:escrow_obligations`	`asset:escrow_partner_holding`
Subscription	`asset:provider_receipts`	`revenue:subscriptions`
Promotion	`asset:provider_receipts`	`revenue:promotions`

7. API architecture

- **Style:** REST under `/api/v1`; JSON; 185 endpoints; OpenAPI in development/test only.
- **Auth:** Bearer access token (15 min) plus an httpOnly refresh cookie scoped to `/api/v1/auth`; CSRF header on cookie-authenticated endpoints.

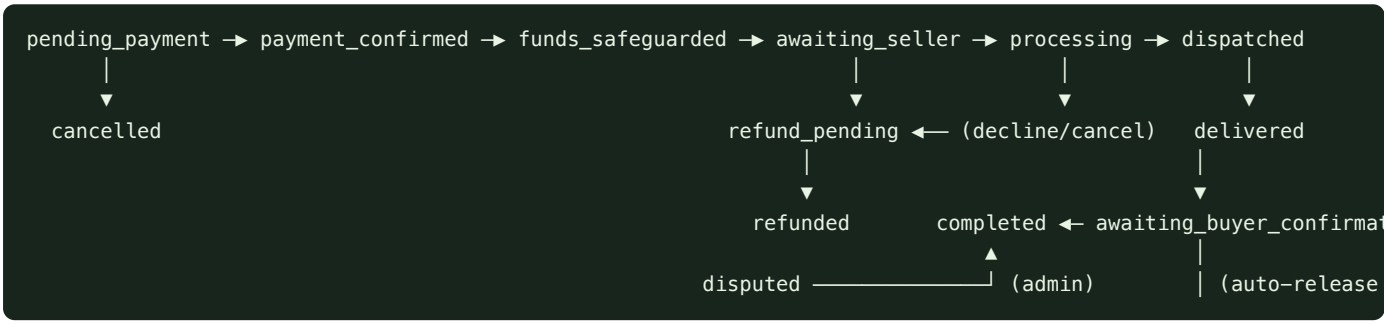
- **Idempotency:** `Idempotency-Key` on checkout; unique `(provider, event_key)` on webhooks; unique ledger keys on settlement.
- **Errors:** consistent `{error:{code,message,details}}`; resource ownership failures return 404 to avoid existence leaks.
- **Rate limits:** per user/device and a coarse per-IP ceiling (20× by default) for carrier NAT; stricter per-action limits on login, OTP, offers, messages, uploads, checkout and Flex.
- **Webhooks:** `POST /api/v1/payments/webhooks/{provider}/webhook`; signatures verified (Paystack HMAC-SHA512, Flutterwave `verif-hash`); invalid signatures return 403; events de-duplicated.
- **Health:** `/health` for liveness, `/health/ready` checks the database and reports providers outside production.

8. Identity and access

Control	Implementation
Passwords	Argon2id; automatic re-hash on login; 8+ characters with letters and digits; timing-safe login for unknown emails.
Access tokens	Short-lived JWT (15 min) kept in memory; algorithm pinned to HS256/384/512; per-user token version revokes all on password/role/status change.
Refresh tokens	Opaque, hashed at rest, rotated on every use; reuse revokes the family; httpOnly, SameSite=Lax, Secure outside development.
OTP	6 digits, HMAC-peppered, 10-minute TTL, 5 attempts, 3 sends/10 min and 10/day per number.
2FA	TOTP with single-use recovery codes and replay protection; required for admins in production; staff cannot use Google sign-in.
Captcha	reCAPTCHA v2 or Turnstile, enforced before the password check for new devices (24 h) or after 3 wrong passwords.
Google sign-in	OIDC web flow; mock consent in development; hidden in production when credentials are absent.
RBAC	Granular admin permissions; only super admins change roles; every admin route has a permission dependency; audit log is append-only.
Proxy trust	`X-Forwarded-For` ignored unless the peer is in `TRUSTED_PROXY_CIDRS`; chain walked right-to-left.

9. Payments, escrow and ledger

Goods.ng does not hold funds. A licensed escrow/payment partner holds them; Goods.ng instructs release or refund based on the order state machine. The ledger mirrors the partner's position.



- **Verify before trust:** server-to-server verify with exact amount and currency; webhook signature; redirects never trusted.
- **No double settlement:** state machine + row locks + unique `escrow:<id>:settle` ledger key; a five-thread concurrency test asserts one settlement.
- **Server-side pricing:** totals from the listing or an accepted negotiation owned by the buyer; DB check constraint on the total.
- **Stock:** reserved atomically at checkout; restored on unpaid expiry or pre-dispatch refund; late payment honoured if stock remains, otherwise auto-refunded.
- **Refunds:** automatic pre-dispatch for cancellation/decline; dispute refunds need a second, different admin; one active refund per order.
- **Reconciliation:** `/admin/reconciliation` cross-checks payments, escrow states and ledger postings.

10. Goods Flex

Flex is a credit product, not a wallet. Before applying, the customer builds a credit profile: verified phone → BVN (Fernet-encrypted, keyed fingerprint to prevent reuse) → read-only bank connection → mandatory guarantor consent → optional workplace verification. Without a guarantor, the limit is zero and no application can start.

Step	Rule
Bank assessment	Six months of transactions; salary detection or 70% of median eligible inflow; deduct existing repayments; risk factor from volatility, missing months, betting share and failed debits.
Limit	`min(statement limit, verification-tier cap, repayment-history cap, lender maximum)`; zero without phone + BVN + bank + consented guarantor.
Guarantor	Sees borrower, relationship, amount and expiry; explicit consent; consent time, IP and BVN reference recorded; amount caps the limit; at most 3 active guarantees per person; withdrawal drops the limit to zero.
Decision	Partner decision stored with reference; offer terms stored verbatim; buyer explicitly accepts.
One plan	Only one of approved/accepted/active/defaulted at a time, enforced in the service under a row lock and by a partial unique index.
Funds	Down payment via the payment layer; partner disburses; order funded through the same escrow path; refund cancels the agreement.
Repayment	Instalments via the payment layer; worker marks overdue and reminds; defaulted plans can still be paid off.

11. Search and discovery

- **Engine:** PostgreSQL full-text search; generated weighted `tsvector` (title A, brand/model B, description C) with a GIN index and prefix matching (`iph` matches iPhone).
- **Ranking:** featured first, then premium sellers, then text rank, then recency.
- **Filters/facets:** state, category, price, condition, brand, BNPL, delivery, verification; counts computed for the current result set.
- **Sponsored:** premium sellers and Boost Premium listings enter a quality-gated sponsored pool; TOP promotes a single listing.
- **Seam:** `products.service.search()` is the single place to swap in a dedicated engine later.
- **Recommendations:** interest signals and a "for you" feed; purge job keeps signal tables bounded.

12. Messaging and negotiation

Area	Behaviour
Conversations	Product-linked; one thread per buyer/seller/product context; read receipts and unread counts; 5-second polling while visible, paused when hidden.
Safety	Direct-contact warning gate; message reporting; rate limits; HTML stripped.
Negotiation	One live negotiation per buyer/product; turn-taking; counter direction; minimum offer 10% of list; 48-hour offer expiry; 24-hour checkout after acceptance; 10-round cap.
Concurrency	Every response locks the negotiation row and must name the pending offer; stale/double clicks get `409 stale_offer`.
Price integrity	The public listing price never changes; the accepted offer becomes the checkout price for that buyer only.

13. Frontend architecture

- **Public pages** (home, search, category, item, store, info) are server-rendered with streaming and PPR; static shells are prerendered; JSON-LD, canonical URLs, OG metadata, sitemap and robots are included.
- **Dashboards** (account, orders, messages, seller, admin) are client components on a central API client: in-memory access token, silent refresh through the httpOnly cookie with a CSRF header, one retry on 401.

- **Performance:** system fonts, backend WebP thumbnails (400 px) and full images (1280 px), lazy loading, skeletons, Redis-cached feeds, bottom navigation on phones.
- **Admin host:** `frontend/src/proxy.ts` serves the console on `admin.goods.ng`, redirects `/admin/...`, and adds `X-Robots-Tag: noindex`. Sessions are per host.
- **Resilience:** no client-side price trust, no direct provider secrets, and no analytics scripts before marketing consent.

14. Media pipeline

1. Input capped at `MAX_UPLOAD_MB` (8 MB) and 40 MP (decompression-bomb guard).
2. Decoded with Pillow; extension and MIME are not trusted.
3. Re-encoded to WebP: full at 1280 px, thumbnail at 400 px; EXIF/GPS and embedded payloads stripped.
4. Listing photos watermarked with the seller business name and Goods.ng.
5. Stored under random keys; public media to S3 + CDN; KYC documents to the private prefix with server-side encryption and short-lived signed URLs.
6. Per-seller cap `MAX_IMAGES_PER_SELLER` (200); soft-deleted listings purged after 30 days.

15. Background processing

Job	Interval	Purpose
`outbox`	10 s	Drain queued email/SMS notifications.
`expire_unpaid_orders`	60 s	Expire unpaid orders, restore stock, re-check late payments with the provider.
`expire_negotiations`	60 s	Expire stale offers and accepted-offer checkout windows.
`auto_release_escrow`	300 s	Release funds after the buyer-confirmation window.
`expire_subscriptions`	300 s	Expire plans, clear promoted tiers and features.
`refresh_promotion_bumps`	3600 s	Refresh Boost Premium top placement.
`bnpl_reminders`	3600 s	Mark overdue instalments and send reminders.
`meta_capi_purchases`	300 s	Send consented purchase events to Meta Conversions API.
`purge_interest_signals`	86400 s	Keep recommendation signals bounded.
`purge_deleted_listings`	86400 s	Hard-delete listings and media after retention.

16. Caching and performance

- **Cache:** Redis with an in-memory development fallback; anonymous search cached for 15 s; public feeds cached; single-flight locks prevent stampedes.
- **Rate limiting:** per user/device and a coarse per-IP ceiling; hot paths use constant-time checks.
- **Database:** 15 s statement timeout, 60 s idle-in-transaction timeout, pooled connections, indexes for search and browse.
- **Measurement:** k6 realistic mix; single dev worker plateaus at ~182 req/s. Add workers/replicas and PgBouncer before that becomes the ceiling.

Scenario	Throughput	p95	Errors
Home feed smoke (20 concurrent)	~300 req/s	~220 ms	0
Search smoke (20 concurrent)	~240 req/s	~125 ms	0
k6 mix, 120 VUs	—	41 ms	0
k6 mix, 400 VUs	182 req/s	553 ms	0
k6 mix, 800 VUs	~182 req/s (plateau)	~1.9 s	0

17. Security architecture

Area	Control
Transport & headers	TLS everywhere; HSTS in staging/production; CSP, nosniff, frame DENY, referrer policy, Permissions-Policy, COOP; `no-store` on auth/admin responses; no Server header in production.
Secrets	Secret manager only; production boot rejects dev defaults, short JWT secrets, http origins, missing Redis, local storage, mock financial providers, missing proxy CIDRs and admin 2FA off.
Input/output	Pydantic validation; HTML stripped with nh3; React escaping; JSON-LD escaped; user URLs restricted to http(s); parameterised queries only.
Files	Decode-and-re-encode; random keys; private KYC prefix; signed URLs; malware scanning still outstanding.
Financial	Append-only ledger/escrow/audit; balance check at commit; row locks; idempotency; four-eyes refunds; reconciliation.
Privacy	KYC identifiers encrypted; masked in API; admin reveal audited; marketing cookies gated; NDPA workflows partially outstanding.
Audit	Every staff sign-in/out/failure, 2FA failure, admin action, permission refusal and console page view; CSV export; immutable rows.
Outstanding	Malware scanning, licensed KYC provider, external pen test, DPO/data-subject workflow, NDPA record of processing.

18. Observability and operations

- **Logs:** structured JSON on stdout with `request_id` on every line and in the `x-request-id` response header; ship to a log platform.
- **Health:** `/health` (liveness) and `/health/ready` (DB, cache, providers outside production).
- **Error tracking:** Sentry SDK for FastAPI and Next.js is **not wired yet**; add DSNs via secrets and extend CSP allowlists.
- **Alerting:** 5xx rate, p95 latency, readiness failures, worker job failures, payment amount mismatch, invalid webhook signatures, non-empty reconciliation, overdue BNPL growth, guarantor limit spikes, provider errors, bank-connection drop.
- **Backups:** managed PITR 7–35 days plus encrypted daily logical dumps to a separate account/region; quarterly restore drill into staging and ledger reconciliation.
- **Rollback:** redeploy the previous immutable image tag; prefer forward database fixes; never blindly downgrade financial tables.

19. Integration architecture and go-live readiness

This section is the architecture-level view of the integration gaps. Appendix A is the detailed checklist with owners and acceptance criteria.

Integration	Interface / state today	Go-live	What remains
Escrow partner	`EscrowProvider`; mock only	BLOCKER	Licensed contract; implement create_hold/release/refund; sandbox certification; payout API; webhooks; reconciliation.
Lending partner (Flex)	`FinancingProvider`; mock only	BLOCKER	Licensed lender; decision/disbursement/repayment APIs; terms; FCCPC review; collections.
Payments	Paystack and Flutterwave adapters; mock active	REQUIRED	Sandbox → live certification; webhook secrets; settlement account; refunds; reconciliation.
Open banking	Mono adapter; mock active	REQUIRED	Mono sandbox test; Connect widget keys; production approval; connection webhooks.
KYC / BVN	`IdentityProvider`; mock only	REQUIRED	Licensed provider adapter (Dojah/Prembly/Smile ID/Youverify); consent; encryption; webhook; manual fallback.

Integration	Interface / state today	Go-live	What remains
SMS	Termii adapter; console active	REQUIRED	Sender ID approval; templates; delivery reports; DND; sandbox OTP test.
Email	Postmark/SMTP adapters; console active	REQUIRED	Domain + SPF/DKIM/DMARC; templates; bounce/complaint handling; sandbox test.
Object storage / CDN	Local + S3 adapter; local active	REQUIRED	Bucket, IAM, CORS, CDN, lifecycle, versioning, private KYC prefix, signed URLs.
Captcha	reCAPTCHA/Turnstile; mock active	REQUIRED	Production keys and domains; CSP already allows providers.
Google OAuth	Implemented; mock in dev	CONFIG	OAuth client, consent screen, redirect URIs, publish app.
Redis	Optional in dev; required in prod	BLOCKER	Managed instance, private network, TLS, monitoring.
Sentry / observability	Not wired	REQUIRED	SDKs, DSNs, release tracking, alerts, on-call, log platform, uptime.
Deployment platform	Dockerfiles/Compose; not deployed	BLOCKER	Vercel, container platform, managed Postgres PITR, PgBouncer, DNS/SSL.
Seller payouts	Ledger only (`seller_payable`)	REQUIRED	Partner payout API; bank account verification; payout scheduling; UI; reconciliation.
Logistics / tracking	Manual tracking fields	RECOMMENDED	At least one carrier; labels; tracking webhooks; delivery status mapping.
Malware scanning	Not implemented	REQUIRED	ClamAV/cloud scanner on upload; quarantine; KYC document handling.
Google Merchant / Ads	Feed endpoint implemented	GROWTH	Merchant Center, domain claim, scheduled fetch, Ads link, campaign.
Meta Pixel / Catalog / CAPI	Pixel events, feed and CAPI implemented	GROWTH	Business Manager, catalog, domain verification, pixel ID, CAPI token, campaign.
GA4 / Google Ads tags	Admin settings + consent gating	GROWTH	Property/tag IDs; conversion labels; consent-mode check.
Push notifications	Not implemented	OPTIONAL	FCM/APNs, device tokens, preferences; needed for native apps.
WhatsApp Business API	Deep links only	OPTIONAL	Cloud API, templates, opt-in, handoff to human agents.
Dedicated search	Postgres FTS	SCALE	OpenSearch/Meilisearch behind `service.search()`.
WebSockets / SSE	5-second polling	SCALE	Replace polling if load or latency demands.
Fraud / risk scoring	Rules + verification only	OPTIONAL	Vendor scoring, device fingerprint, velocity rules.

20. Scalability plan

1. **Horizontal API:** add replicas; keep the app stateless; move sessions/tokens as already designed.
2. **Database:** PgBouncer transaction pooling; read replicas for browse/search; partition large event tables if needed; then shard only as a last resort.
3. **Cache:** increase Redis capacity; add per-entity cache invalidation; cache anonymous feeds and facets.
4. **Search:** move to a dedicated engine when Postgres FTS p95 or query complexity becomes the bottleneck.
5. **Messaging:** WebSockets/SSE or a managed pub/sub when polling at scale becomes expensive.
6. **Media:** CDN offload, responsive variants, signed URLs; keep origin egress low.
7. **Worker:** more replicas are safe; split jobs by queue if contention appears.

21. Architecture decisions

ADR	Decision	Rationale
001	Modular monolith, not microservices	Small team, shared transactions, fast iteration; module seams allow later extraction.
002	PostgreSQL as the only source of truth	Strong constraints, transactions, FTS and operational simplicity; ledger integrity is paramount.
003	Provider interfaces for regulated services	Partners change; the domain must not. Production refuses mocks.
004	Same-origin API via Next.js rewrite	First-party refresh cookie, no CORS, simpler CSP, better mobile reliability.
005	Double-entry ledger with DB triggers	Money correctness cannot depend on application discipline alone.
006	Verify-before-trust payments	Webhooks and redirects are signals, not proof.
007	Premium is derived, not stored	An active subscription period is the only truth; clients cannot self-grant.
008	Per-listing promotions separate from subscriptions	Lowers the barrier: a seller can buy visibility for one product without a plan.
009	Postgres FTS first	Sufficient at launch; one seam to replace later.
010	Polling messaging first	Simpler operations; WebSockets can replace without API changes.

22. Technical risks

Risk	Mitigation
Escrow/lending adapter surprises in sandbox	Contract-first adapter tests; certify in staging before production; keep mock for regression tests.
Webhook storms or missed events	Idempotency, de-duplication, buyer-return verification, expiry job re-check, alerting.
Database connection exhaustion	Pool math, PgBouncer, load tests, connection alerts.
Large-image abuse	Size and pixel caps, re-encode, per-seller quota, malware scan.
KYC data breach	Private storage, encryption, signed URLs, least privilege, audit, pen test.
Provider outage	Retries with backoff, status page, manual reconciliation, graceful buyer messaging.
Search relevance drift	Ranking tests, stable weights, search success metric.
Polling load at scale	Visibility-aware polling, backoff, then WebSockets/SSE.

PART III

Functional Requirements Document

Testable behaviour for every module: actors, requirements, business rules, state machines, non-functional targets, integration requirements and end-to-end acceptance scenarios.

1. Conventions and actors
2. Authentication
3. Sellers and verification
4. Catalog and listings
5. Search
6. Messaging and negotiation
7. Payments and checkout
8. Orders and disputes
9. Goods Flex
10. Reviews and notifications
11. Subscriptions and promotions
12. Ads and feeds
13. Administration
14. State machines
15. Non-functional
16. Integration requirements
17. Acceptance scenarios

1. Conventions and actors

Requirements use stable IDs: `FR-<AREA>-<NNN>`. Priority is **P0** (launch blocker), **P1** (launch-critical but degradable), **P2** (post-launch) or **P3** (optional). Each requirement is written so that it can become a test case. "Must" is mandatory; "should" is expected unless a documented exception exists.

Actor	Description	Capabilities
Guest	Unauthenticated visitor.	Browse, search, view listings/stores/reviews, register, sign in.
Buyer	Any authenticated user.	Chat, negotiate, checkout, order, review, favourite, apply for Flex.
Seller	A user with a seller profile.	List, manage stock, answer, accept/decline, dispatch, view analytics, buy promotions.
Premium seller	Seller with an active subscription.	Premium badge, contact buttons, banner, analytics, featured credits, priority ranking.
Guarantor	Person consenting to a Flex guarantee.	View terms, consent/decline, withdraw; no marketplace account required.
Admin	Staff with granular permissions.	Moderate, verify, resolve, refund, reconcile, configure, audit.
Super admin	Staff with all permissions.	Everything, including role changes and staff management.
System	Worker/scheduler.	Expiry, release, reminders, outbox, feeds, CAPI, purge.

Priority definitions

Priority	Meaning
P0	Cannot launch without it; security, money or core journey.
P1	Launch-critical; may launch in a limited mode with a documented workaround.
P2	Post-launch enhancement.
P3	Optional / scale.

2. Authentication and accounts (FR-AUTH)

ID	Requirement	Priority	Acceptance criteria
`FR-AUTH-001`	Register with email, phone, name and password.	P0	Valid input creates account; duplicate email/phone rejected; password policy enforced; response does not leak existence.
`FR-AUTH-002`	Verify phone with a 6-digit OTP.	P0	OTP expires in 10 min; 5 attempts; resend limits 3/10 min and 10/day; success marks phone verified.
`FR-AUTH-003`	Sign in with email + password.	P0	Correct credentials return access + refresh; wrong password increments lockout; 5 failures lock 15 min.
`FR-AUTH-004`	Captcha challenge for new devices or after failures.	P0	Required when no valid device marker within 24 h or after 3 wrong passwords; verified server-side before password check.
`FR-AUTH-005`	Rotating refresh tokens with reuse detection.	P0	Refresh rotates; reuse revokes the family; cookie httpOnly/SameSite=Lax/Secure in prod; CSRF header required.
`FR-AUTH-006`	Password reset without account enumeration.	P0	Same response for known/unknown email; token single-use and expiring; reset revokes sessions.
`FR-AUTH-007`	TOTP 2FA with recovery codes.	P1	Setup returns secret + QR; enable requires a valid code; recovery codes single-use; replay rejected; disable requires password/code.
`FR-AUTH-008`	Google sign-in.	P2	OIDC flow; no staff accounts; 2FA still required; button hidden when not configured.
`FR-AUTH-009`	Session revocation.	P0	Password change, role change, account disable or logout invalidates access via token version and refresh family.
`FR-AUTH-010`	Admin 2FA enforcement.	P0	With `ADMIN_2FA_REQUIRED=true`, staff without 2FA cannot use admin APIs; boot fails if off in production.

3. Sellers and verification (FR-SEL)

ID	Requirement	Priority	Acceptance criteria
FR-SEL-001	Become a seller.	P0	One seller profile per user; store name, slug, state, city, contact; duplicate slug handled.
FR-SEL-002	Storefront and branding.	P1	Public store page with logo, banner, about, listings, reviews and verification badge.
FR-SEL-003	Verification tiers.	P0	Phone → identity → business; documents stored privately; admin decision; tier visible on listings.
FR-SEL-004	BVN lookup and name match.	P1	Encrypted BVN; keyed fingerprint prevents reuse; lenient name match; failure reason recorded.
FR-SEL-005	Seller analytics.	P1	Views, favourites, enquiries, orders, earnings, conversion; premium-only depth gated server-side.
FR-SEL-006	Seller suspension.	P0	Admin can suspend; listings hidden; sessions revoked; audit entry written.
FR-SEL-007	Premium contact controls.	P1	WhatsApp/phone buttons only for active premium; server-enforced; client flags ignored.

4. Catalog and listings (FR-CAT, FR-LST)

ID	Requirement	Priority	Acceptance criteria
FR-CAT-001	Category tree.	P0	17 roots, 91 categories; two levels; slug unique; active/inactive; admin reorder.
FR-CAT-002	Locations.	P0	36 states + FCT; cities per state; listing state must be valid.
FR-LST-001	Create a listing.	P0	Title, category, condition, price, stock, description, location, delivery options; validation errors are specific.
FR-LST-002	Image upload and ordering.	P0	Up to per-seller cap; re-encode to WebP; watermark; reorder; delete; first image is cover.
FR-LST-003	Auto-title suggestions.	P2	Deterministic suggestions from brand/model/condition/category; detects year, storage, colour, size; does not invent facts.
FR-LST-004	Listing lifecycle.	P0	Draft/active/paused/sold/removed; soft delete; purge after 30 days; views and favourites counted.
FR-LST-005	Stock and availability.	P0	Quantity decremented atomically at checkout; restored on expiry/refund; zero-stock blocks checkout.
FR-LST-006	Delivery options.	P0	Pickup, courier, seller delivery; fee and ETA; buyer selects at checkout; server validates the choice.
FR-LST-007	Moderation.	P0	Admin approve/reject with reason; rejected listings hidden; seller notified; audit.
FR-LST-008	Promotion badges.	P1	TOP and Premium badges reflect active promotions only; cleared automatically on expiry.

5. Search and discovery (FR-SRC)

ID	Requirement	Priority	Acceptance criteria
FR-SRC-001	Keyword search.	P0	Prefix matching (e.g. "iph" → iPhone); weighted title/brand/description; empty query returns a browse feed.
FR-SRC-002	Filters and sorting.	P0	State, category, price range, condition, brand, BNPL, delivery; sort by relevance, newest, price.

ID	Requirement	Priority	Acceptance criteria
FR-SRC-003	Facets.	P1	Counts for the current result set; selecting a facet updates results without losing context.
FR-SRC-004	Ranking.	P1	Featured → premium → text rank → recency; no paid placement that fails quality gates.
FR-SRC-005	Favourites and saved items.	P1	Toggle favourite; count on listing; buyer-only list; idempotent.
FR-SRC-006	Recommendations.	P2	"For you" from interest signals; excludes inactive/blocked listings; bounded signal storage.
FR-SRC-007	Cache safety.	P1	Anonymous search cache TTL 15 s; no user-specific data in shared cache; invalidation on listing changes.

6. Messaging and negotiation (FR-MSG, FR-NEG)

ID	Requirement	Priority	Acceptance criteria
FR-MSG-001	Start a product-linked chat.	P0	Buyer and seller only; one thread per context; non-participants get 404; rate limited.
FR-MSG-002	Send and read messages.	P0	Text sanitised; pagination; read receipts; unread counts; polling every 5 s while visible.
FR-MSG-003	Report a message.	P0	Report created; admin queue; reporter confirmation; no retaliation exposure.
FR-MSG-004	Direct-contact warning.	P0	Buyer sees a warning before off-platform contact; acknowledgement recorded; premium contact still gated.
FR-NEG-001	Make an offer.	P0	Offer ≥ 10% of list price; one live negotiation per buyer/product; 48 h expiry; seller notified.
FR-NEG-002	Counter-offer.	P0	Turn-taking enforced; counter direction enforced; 10-round cap; stale offer ID returns 409.
FR-NEG-003	Accept and checkout.	P0	Accepted offer locks the price for that buyer for 24 h; checkout uses the agreed price; listing price unchanged.
FR-NEG-004	Expiry and withdrawal.	P0	Expired offers cannot be accepted; withdrawal notifies the other party; expired negotiation frees the slot.

7. Payments and protected checkout (FR-PAY, FR-CHK)

ID	Requirement	Priority	Acceptance criteria
FR-PAY-001	Provider abstraction.	P0	Mock, Paystack and Flutterwave implement initialize/verify/refund/webhook; active provider selected by env.
FR-PAY-002	Server-side verification.	P0	Success requires exact amount and currency from the provider; underpayment never funds an order.
FR-PAY-003	Webhook security.	P0	Signature verified; invalid → 403; events de-duplicated by `(provider, event_key)`; webhook only triggers verification.
FR-PAY-004	Idempotent checkout.	P0	Same `Idempotency-Key` returns the same payment/order; duplicate clicks do not double-charge.
FR-PAY-005	Refunds.	P0	Full or amount-based refund via provider; one active refund per order; status reconciled.
FR-CHK-001	Checkout quote.	P0	Subtotal + delivery + protection fee = total; server-computed; protection capped; DB constraint enforces the sum.

ID	Requirement	Priority	Acceptance criteria
FR-CHK-002	Create order and reserve stock.	P0	Atomic stock decrement; failure returns a clear out-of-stock error; unpaid order expires in 60 min.
FR-CHK-003	Escrow hold.	P0	On verified payment, partner creates a hold; funds are not released until the order state allows it.
FR-CHK-004	Late payment.	P0	After expiry, re-check provider; honour if stock remains, otherwise auto-refund.
FR-CHK-005	Reconciliation.	P0	Admin report cross-checks payments, escrow and ledger; mismatches are surfaced, not silently corrected.

8. Orders, delivery, disputes and refunds (FR-ORD, FR-DSP)

ID	Requirement	Priority	Acceptance criteria
FR-ORD-001	Seller accepts or declines.	P0	Only the seller can act; decline triggers refund; accept moves to processing.
FR-ORD-002	Dispatch and tracking.	P0	Carrier, tracking number/URL and updates recorded; buyer notified; status order enforced.
FR-ORD-003	Delivery evidence.	P0	Photos/notes attached; visible to buyer/admin; audit trail.
FR-ORD-004	Buyer confirmation and auto-release.	P0	Buyer confirms → completed; otherwise auto-release after 3 days; no release while disputed.
FR-ORD-005	Cancellation.	P0	Pre-dispatch cancellation refunds automatically; post-dispatch requires a dispute.
FR-DSP-001	Open a dispute.	P0	Buyer opens within the allowed window; reason + evidence; seller notified; order frozen.
FR-DSP-002	Dispute conversation.	P0	Buyer/seller/admin messages; evidence; timeline; no participant can delete history.
FR-DSP-003	Resolution.	P0	Admin resolves with release or refund; refund needs a second, different admin; audit entry.
FR-DSP-004	Partial refunds.	P2	Model supports amounts; UI and flow to be added post-launch.

9. Goods Flex (FR-FLX)

ID	Requirement	Priority	Acceptance criteria
FR-FLX-001	Credit profile.	P1	Verified phone, BVN, connected bank, consented guarantor, optional employment; missing any required step leaves the limit at zero.
FR-FLX-002	Affordability assessment.	P1	Six months classified; income detected; repayments deducted; risk factor applied; snapshot stored immutably with expiry.
FR-FLX-003	Guarantor consent.	P1	SMS link; borrower, relationship, amount, expiry shown; explicit consent; IP/time/BVN ref recorded; decline/withdraw supported; max 3 active.
FR-FLX-004	Apply and accept.	P1	Partner decision stored; terms shown verbatim; explicit acceptance; unpaid offer expires in 24 h.
FR-FLX-005	One running plan.	P1	Approved/accepted/active/defaulted blocks a new plan; enforced under lock and by partial unique index.
FR-FLX-006	Down payment and disbursement.	P1	Down payment via payment layer; partner disburses; order funded through escrow; refund cancels the agreement.
FR-FLX-007	Instalments.	P1	Schedule shown; pay via payment layer; duplicate/excess payments auto-refunded; one open payment per instalment.

ID	Requirement	Priority	Acceptance criteria
FR-FLX-008	Overdue handling.	P1	Worker marks overdue and reminds; defaulted plans can still be paid off; no double reminders.

10. Reviews, notifications and trust (FR-REV, FR-NOT, FR-TRS)

ID	Requirement	Priority	Acceptance criteria
FR-REV-001	Review eligibility.	P1	Only a buyer with a completed/delivered order can review; one review per order/item.
FR-REV-002	Rating and content.	P1	1–5 stars; text sanitised; seller reply; report; admin moderation.
FR-REV-003	Aggregates.	P1	Seller rating average/count updated; no review hidden without a moderation record.
FR-NOT-001	In-app inbox.	P0	Unread counts; read/unread; read-all; pagination; per-user isolation.
FR-NOT-002	Preferences.	P1	Per-channel opt-in; transactional messages still delivered; unsubscribe respected.
FR-NOT-003	Email/SMS outbox.	P0	Queued transactionally; worker retries; console provider in dev; Termii/Postmark/SMTP in production.
FR-TRS-001	Listing moderation.	P0	Reported listings queued; take down with reason; repeat offenders suspended.
FR-TRS-002	Prohibited items.	P0	Policy page; report flow; admin removal; audit.
FR-TRS-003	Account suspension.	P0	Immediate session revocation; listings hidden; appeal path recorded.

11. Subscriptions and promotions (FR-SUB, FR-PRO)

ID	Requirement	Priority	Acceptance criteria
FR-SUB-001	Plans.	P1	Monthly ₦15,000/30d/3 credits; Quarterly ₦40,000/90d/10; Annual ₦150,000/365d/50; admin can edit.
FR-SUB-002	Purchase and activation.	P1	Payment verified server-side; period set; credits granted; premium derived from active period.
FR-SUB-003	Featured credits.	P1	Spend a credit to feature a listing; balance decremented atomically; expiry respected.
FR-SUB-004	Admin grant/cancel.	P1	Grant creates an active period; cancel sets end; both audited.
FR-PRO-001	Packages.	P1	TOP 7 days ₦3,099; TOP 30 days ₦9,999; Boost Premium 28 days ₦31,999; server-priced.
FR-PRO-002	Purchase.	P1	Creates a promotion payment; activation only after verified success; idempotent.
FR-PRO-003	Eligibility.	P1	Boost Premium requires verified seller, 3+ photos and 60+ char description; TOP has no such gate.
FR-PRO-004	Expiry.	P1	Worker clears tier/featured after the window; badges disappear; no stale premium placement.
FR-PRO-005	Extend.	P2	Seller can extend an active promotion; price and window are server-computed.

12. Ads, feeds and analytics (FR-ADS)

ID	Requirement	Priority	Acceptance criteria
FR-ADS-001	Google product feed.	P2	XML at `/feeds/google.xml`; only Premium listings with verified seller, 3+ photos and real description; property/vehicles/jobs/services excluded by default.
FR-ADS-002	Meta catalog feed.	P2	CSV at `/feeds/meta.csv`; same eligibility.
FR-ADS-003	Optional feed token.	P3	When `FEED_TOKEN` is set, requests without it are rejected.
FR-ADS-004	Meta CAPI purchases.	P2	Only consented buyers; emails/phones SHA-256 hashed; worker retries; token from env only.
FR-ADS-005	Consent gating.	P0	No marketing scripts before consent; cookie settings available in footer; NDPA copy.

13. Administration (FR-ADM)

ID	Requirement	Priority	Acceptance criteria
FR-ADM-001	Granular permissions.	P0	Every admin route requires a permission; missing permission returns 403 and is audited.
FR-ADM-002	Staff management.	P0	Invite by email; no self-signup; role assignment only by super admin; invite expires.
FR-ADM-003	Audit log.	P0	Append-only; actor/IP/device; filters; CSV export; export itself audited.
FR-ADM-004	Platform settings.	P0	Fees, caps, moderation switch, feeds, tracking IDs, promotion packages; validated and audited.
FR-ADM-005	Reconciliation.	P0	Payments vs escrow vs ledger; mismatch list; no silent mutation.
FR-ADM-006	Moderation queues.	P0	Listings, reviews, reports, verifications, disputes, refunds; SLA fields; assignment.
FR-ADM-007	Admin subdomain.	P1	`admin.goods.ng`; redirects; noindex; separate session; optional IP allow-list.
FR-ADM-008	First super admin.	P0	CLI creates the first account with a one-time invite; refuses if an active super admin exists unless forced.

14. State machines

14.1 Order / escrow



Only `modules/orders/escrow.py` changes these statuses. Release/refund are guarded by row locks, a unique ledger key and the state machine.

14.2 Payment

State	Meaning	Allowed next
initialized	Provider checkout created; not trusted.	pending, failed, expired
pending	Awaiting provider confirmation.	paid, failed, expired
paid	Server-to-server verified with exact amount/currency.	refunded
failed / expired	No funds captured; order may expire.	—
refunded	Refund confirmed by provider.	—

14.3 Negotiation

State	Meaning	Allowed next
open	Offer/counter pending; turn owner defined.	accepted, declined, expired, withdrawn, converted
accepted	Price agreed; 24 h checkout window.	converted, expired
converted	Order paid at the agreed price.	—
declined / withdrawn / expired	No live offer; slot freed.	—

14.4 Goods Flex plan

State	Meaning	Allowed next
approved	Lender approved; terms offered.	accepted, expired, withdrawn
accepted	Buyer accepted; awaiting down payment/disbursement.	active, cancelled
active	Disbursed; instalments due.	completed, defaulted, cancelled
defaulted	Missed instalments; still repayable.	completed, cancelled
completed / cancelled	Plan closed; slot freed.	—

15. Non-functional requirements

ID	Category	Requirement	Verification
NFR-PERF-001	Performance	Public read endpoints p95 < 500 ms at target concurrency; home and search cached.	k6 + smoke test.
NFR-PERF-002	Performance	Images served as WebP; thumbnails 400 px; full 1280 px; lazy loaded.	Lighthouse/visual check.
NFR-AVL-001	Availability	API and worker run ≥ 2 / ≥ 1 replicas; readiness checks gate traffic.	Deploy topology review.
NFR-AVL-002	Recovery	PITR enabled; daily logical dump; quarterly restore drill; ledger reconciliation after restore.	Runbook + drill record.
NFR-SEC-001	Security	All money transitions row-locked and idempotent; ledger balanced at commit.	Concurrency tests + DB triggers.
NFR-SEC-002	Security	Production refuses dev secrets and mock financial providers.	Boot-guard tests.
NFR-SEC-003	Security	KYC data encrypted at rest and private; access audited.	Storage/API tests + review.
NFR-PRV-001	Privacy	Consent before marketing scripts; data-subject export/delete workflow; retention schedule.	NDPA review.
NFR-CMP-001	Compliance	Escrow/lending only through licensed partners; disclosures reviewed.	Legal sign-off.
NFR-A11Y-001	Accessibility	Keyboard operable, visible focus, labelled controls, colour contrast on core journeys.	Manual audit + axe.
NFR-I18N-001	Localisation	NGN formatting, Nigerian states/cities, English at launch.	Unit + E2E.
NFR-OBS-001	Observability	Request ID on every log line; health/readiness; error tracking; alerting.	Runbook + dashboards.
NFR-OPS-001	Operations	Migrations backward-compatible; release job; rollback image tag; no prod downgrade on financial tables.	Release checklist.

16. Integration requirements

Each integration requirement is a launch gate. The detailed checklist is in Appendix A.

ID	Integration	Requirement	Priority
IR-01	Escrow	Licensed partner adapter implements hold/release/refund, webhooks and reconciliation; certified in sandbox.	P0
IR-02	Lending	Licensed lender adapter implements decision, disbursement and repayment; FCCPC review complete.	P0
IR-03	Payments	Paystack or Flutterwave certified; webhook secret; settlement; refunds; reconciliation.	P0
IR-04	Open banking	Mono sandbox and production approval; Connect widget; read-only consent; connection events.	P1
IR-05	KYC	Licensed BVN/NIN provider; consent; encrypted storage; admin reveal audited.	P0
IR-06	SMS	Termii sender ID approved; OTP and transactional templates; delivery reports.	P0
IR-07	Email	Postmark/SMTP with SPF/DKIM/DMARC; templates; bounce handling.	P0
IR-08	Storage	S3 bucket + CDN; private KYC prefix; signed URLs; lifecycle.	P0
IR-09	Redis	Managed Redis in production; TLS; monitoring.	P0
IR-10	Captcha	reCAPTCHA/Turnstile production keys; server-side verification.	P0
IR-11	Observability	Sentry, log platform, uptime, alerts, on-call.	P0
IR-12	Deployment	Vercel + containers + managed Postgres/Redis + DNS/SSL.	P0
IR-13	Payouts	Partner payout API; bank account verification; scheduling; reconciliation UI.	P1
IR-14	Logistics	At least one carrier; labels; tracking webhooks.	P2
IR-15	Malware scan	Upload scanning and quarantine for KYC and listing images.	P0
IR-16	Google Ads	Merchant Center + Ads campaign; feed scheduled; domain claimed.	P2
IR-17	Meta Ads	Business Manager, catalog, pixel, CAPI token, campaign.	P2
IR-18	Push	FCM/APNs device tokens and preferences.	P3

17. Acceptance scenarios

#	Scenario	Expected result
AT-01	New user registers, verifies phone, signs in.	Account created; OTP verified; session issued; no sensitive data in response.
AT-02	Seller creates a listing with 3 photos and a 60+ char description.	Listing active; images watermarked and re-encoded; searchable; promotion eligibility true.
AT-03	Buyer makes an offer, seller counters, buyer accepts.	Turn-taking enforced; accepted price locked for 24 h; listing price unchanged.
AT-04	Buyer pays through the provider (sandbox).	Payment verified server-to-server; escrow hold created; stock decremented; seller notified.
AT-05	Duplicate webhook arrives five times.	Exactly one payment confirmation and one settlement; no double credit.
AT-06	Wrong amount is sent.	Payment rejected/underpayment not funded; order remains unpaid/cancelled.
AT-07	Seller dispatches with evidence; buyer confirms.	Order completed; ledger release posts commission and protection fee; seller payable recorded.
AT-08	Buyer disputes before release.	Order frozen; no auto-release; admin resolves; refund requires a second admin.
AT-09	Flex customer connects bank, names guarantor, guarantor consents, applies.	Limit computed; one plan; down payment; disbursement; instalments scheduled.
AT-10	Second Flex application while a plan is running.	Rejected with a clear error; partial unique index prevents a race.

#	Scenario	Expected result
AT-11	Seller buys TOP 7 and Boost Premium.	Server-priced; payment verified; badges and placement active; expiry clears them.
AT-12	Admin without permission calls an admin endpoint.	403; permission refusal audited; no state change.
AT-13	Production boot with mock escrow/BNPL or dev secrets.	Process refuses to start with a clear configuration error.
AT-14	Anonymous search is cached; a listing changes.	Cache TTL bounded; stale result is acceptable for ≤ 15 s; no user data leaks.
AT-15	Marketing consent is declined.	No pixel/GA scripts load; transactional email/SMS still works; consent can be changed.

Definition of done for a requirement
Code + tests (unit/integration/E2E where relevant) + migration if data changes + admin/ops visibility + documentation update. P0 requirements also need a staging verification against the real provider sandbox where an integration is involved.

APPENDIX

Go-Live Integration Checklist & Reference

The external services, credentials, contracts and operational controls that remain between the current development build and a live, licensed marketplace.

A. Integration go-live checklist

B. Environment variables

C. Glossary

A. Integration go-live checklist

Status vocabulary: LIVE implemented and tested against mocks/local; CONFIG code exists but production credentials/dashboard work remains; MOCK only a mock adapter exists; NOT BUILT no implementation. "Blocker" means the item must be complete before the first live transaction or before the production process will boot.

#	Integration	Why it matters	Current state	What remains	Blocker	Owner
1	**Licensed escrow / payment partner**	Holds buyer funds for Protected Checkout; Goods.ng must not hold funds.	MOCK EscrowProvider only	Contract; implement hold/release/refund; sandbox certification; settlement/payout API; webhooks; reconciliation.	Yes	Finance + Eng
2	**Licensed BNPL lender**	Provides the credit decision, disbursement and collections for Goods Flex.	MOCK FinancingProvider only	Contract; decision/disbursement/repayment adapter; real terms; FCCPC review; collections process.	Yes	Risk + Legal
3	**Payment provider** — Paystack or Flutterwave	Collects card/transfer/bank payments and verifies them server-side.	CONFIG both adapters written, untested	Sandbox account; certify initialize/verify/refund; webhook secret and URL; settlement account; reconciliation; choose primary/backup.	Yes	Finance + Eng
4	**KYC / BVN-NIN provider**	Verifies seller identity and Flex borrowers; supports the "Verified" badge at scale.	MOCK IdentityProvider only	Choose Dojah/Prembly/Smile ID/Youverify; adapter; consent; encryption; webhook; manual review fallback; NDPA terms.	Yes	Compliance + Eng
5	**Open banking** — Mono	Read-only bank statements for the Flex affordability assessment.	CONFIG Mono adapter, untested	Mono sandbox; Connect widget keys; production approval; connection webhooks; error monitoring; consent copy.	No*	Risk + Eng
6	**SMS** — Termii	OTP, guarantor consent links and transactional alerts.	CONFIG adapter, console active	Sender ID approval; templates; DND handling; delivery reports; sandbox OTP proof; spend limits.	Yes	Eng + Ops
7	**Email** — Postmark or SMTP	Verification, receipts, dispute and Flex notifications.	CONFIG adapters, console active	Domain verification; SPF/DKIM/DMARC; templates; bounce/complaint webhooks; sandbox test.	Yes	Eng + Ops
8	**Object storage + CDN**	Durable public listing media and private KYC documents.	CONFIG S3 adapter, local active	Bucket; IAM least privilege; CORS; CDN; versioning; lifecycle; private prefix; signed URLs; encryption.	Yes	Platform
9	**Managed Redis**	Shared cache, rate limits and single-flight locks.	CONFIG optional in dev	Provision with TLS/private network; monitoring; eviction policy; boot check in production.	Yes	Platform
10	**Captcha** — reCAPTCHA v2 or Turnstile	Bot protection on sign-in for new devices and after failures.	CONFIG providers implemented, mock active	Production keys; allowed domains; CSP already allows; verify server-side in staging.	Yes	Eng
11	**Sentry / error tracking**	Detect and triage production errors in API and frontend.	NOT BUILT	Add FastAPI + Next.js SDKs; DSNs in secrets; release tags; source maps; alert routing; CSP updates.	Yes	Platform
12	**Deployment platforms**	Run the API, worker, frontend, database and cache in production.	NOT BUILT Dockerfiles/Compose only	Vercel project; container platform; ≥2 API instances; worker; managed Postgres PITR; PgBouncer; DNS/SSL.	Yes	Platform

#	Integration	Why it matters	Current state	What remains	Blocker	Owner
13	**Seller payouts / disbursement**	Move released seller balances out of the escrow partner to seller bank accounts.	NOT BUILT ledger only	Partner payout API; bank account verification/name lookup; payout scheduling; retry/failed handling; UI; reconciliation.	Yes	Finance + Eng
14	**Malware scanning**	Scan listing images and KYC documents before storage/access.	NOT BUILT	ClamAV or cloud scanner; quarantine bucket; scan status; admin review; alerting.	Yes	Security
15	**Logistics / carrier**	Labels, pickup and tracking events for dispatched orders.	NOT BUILT manual tracking fields	At least one carrier (GIGL, DHL, Sendbox, etc.); label API; tracking webhooks; status mapping; failed-delivery flow.	No	Ops + Eng
16	**Google Merchant Center + Ads**	Product listing ads and Shopping/Performance Max campaigns.	CONFIG feed endpoint	Merchant Center; claim goods.ng; Nigeria/NGN; scheduled fetch; link Ads; campaign; feed quality checks.	No	Growth
17	**Meta Pixel / Catalog / CAPI**	Catalog ads and consented server-side purchase events.	CONFIG pixel, feed, CAPI code	Business Manager; catalog data source; domain verification; pixel ID; CAPI token; campaign; event dedup check.	No	Growth
18	**GA4 / Google Ads tags**	Acquisition and conversion measurement.	CONFIG settings + consent gate	Property/tag IDs; conversion labels; consent mode; verify no scripts before consent.	No	Growth
19	**Google OAuth**	Optional social sign-in.	CONFIG implemented, mock in dev	OAuth web client; consent screen; redirect URIs; publish/verify app; privacy/terms links.	No	Eng
20	**Push notifications**	Re-engage buyers/sellers; required for future native apps.	NOT BUILT	FCM/APNs; device tokens; preferences; deep links; delivery analytics.	No	Eng
21	**WhatsApp Business API**	Structured messaging and template notifications.	NOT BUILT deep links only	Cloud API; number; templates; opt-in; webhook; human handoff; 24-hour window handling.	No	Growth + Eng
22	**Dedicated search**	Search relevance and latency at scale.	NOT BUILT Postgres FTS	OpenSearch/Meilisearch; index pipeline; ranking parity; A/B; swap behind `service.search()`.	No	Eng
23	**WebSockets / SSE**	Lower-latency messaging if polling becomes costly.	NOT BUILT 5 s polling	Gateway; auth; fan-out; presence; reconnect; load test; mobile background behaviour.	No	Eng
24	**Fraud / risk scoring**	Detect abusive signups, listings and payments.	NOT BUILT rules only	Vendor or in-house scoring; device fingerprint; velocity rules; manual review queue; model governance.	No	Trust & Safety
25	**NDPA / DPO tooling**	Data-subject export/delete, retention and processing records.	PARTIAL copy only	DSR workflow; retention jobs; record of processing; DPO appointment; processor agreements; breach playbook.	Yes	Compliance

* Mono is not a production boot blocker, but Goods Flex cannot go live without it (or another open-banking provider) because the credit limit depends on a real bank assessment. If Flex is feature-flagged off at launch, Mono can follow.

Top five blockers, in order

1. Licensed escrow/payment partner (Protected Checkout cannot launch without it). **2.** Licensed lender (Goods Flex cannot launch without it). **3.** Payment provider certification (no live money). **4.** Production deployment + managed infrastructure (nothing is live). **5.** KYC + SMS/email + captcha (security and account recovery).

B. Environment variables

Variable	Purpose / production requirement
`ENVIRONMENT`	development | test | staging | production. Controls boot guards and dev routes.
`DATABASE_URL`	PostgreSQL DSN; must use sslmode=require/verify-full in production.
`DB_POOL_SIZE / DB_MAX_OVERFLOW / DB_POOL_TIMEOUT`	Per-process pool; size against Postgres max_connections or PgBouncer.
`REDIS_URL`	Required in production; shared cache, rate limits, single-flight.
`JWT_SECRET / OTP_SECRET / DATA_ENCRYPTION_KEY`	Unique per environment; production rejects dev defaults; back up the Fernet key.
`FRONTEND_URL / CORS_ORIGINS / ADMIN_URL`	Explicit https origins; admin subdomain used in staff invites.
`TRUSTED_PROXY_CIDRS`	Load balancer CIDRs; required in production for correct client IPs.
`ADMIN_2FA_REQUIRED`	Must be true in production; admins enable TOTP before using the admin API.
`PAYMENT_PROVIDER`	mock | paystack | flutterwave; mock refused in production.
`ESCROW_PROVIDER / BNPL_PROVIDER`	Mock only today; production refuses mock; real adapters required.
`BANK_DATA_PROVIDER / MONO_*`	mock | mono; Mono keys server-side only; production refuses mock.
`PAYSTACK_SECRET_KEY / FLUTTERWAVE_*`	Provider credentials and webhook hash.
`SMS_PROVIDER / TERMII_*`	console | termii; production refuses console.
`EMAIL_PROVIDER / SMTP_* / POSTMARK_*`	console | smtp | postmark; production refuses console.
`STORAGE_BACKEND / S3_* / MEDIA_BASE_URL`	local | s3; production requires s3 + bucket; CDN URL for media.
`CAPTCHA_PROVIDER / *_SITE_KEY / *_SECRET_KEY`	off | mock | recaptcha | turnstile; production refuses mock/off.
`GOOGLE_CLIENT_ID / SECRET`	Google OAuth; hidden in production when absent.
`META_CAPI_ACCESS_TOKEN / META_GRAPH_VERSION`	Server-side purchase events; token never in DB/client.
`FEED_TOKEN`	Optional protection for Google/Meta feeds.
`PLATFORM_COMMISSION_RATE / PROTECTION_FEE_RATE / PROTECTION_FEE_CAP`	Marketplace economics; admin-overridable.
`AUTO_RELEASE_DAYS`	Buyer-confirmation window before escrow release (default 3).
`RATE_LIMIT_PER_MINUTE / RATE_LIMIT_IP_MULTIPLIER`	Per-identity and coarse per-IP ceilings.
`MAX_UPLOAD_MB / MAX_IMAGES_PER_SELLER / DELETED_LISTING_RETENTION_DAYS`	Media abuse and retention controls.

Variable	Purpose / production requirement
`RATE_LIMIT_MULTIPLIER`	Dev/test only; must be 1 outside development.

C. Glossary

Term	Meaning
Protected Checkout	Goods.ng checkout where a licensed partner holds the buyer's funds until delivery is confirmed or the dispute process resolves.
Escrow partner	A licensed third party that holds and releases funds. Goods.ng does not hold funds.
Goods Flex	Buy-now-pay-later product with an affordability assessment, mandatory guarantor and instalments.
Guarantor	A person who consents to cover a Flex borrower's obligation up to a stated, capped amount.
Premium seller	Derived status from an active paid subscription; unlocks contact buttons, analytics, banner and ranking.
Promotion	A one-off paid visibility package for a single listing: TOP 7, TOP 30 or Boost Premium 28.
Boost Premium	Promotion tier that makes a listing eligible for the sponsored pool while it meets quality gates.
Ledger	Double-entry, append-only record of money movements; balanced at commit by a deferred trigger.
Seller payable	Ledger liability representing money owed to a seller after release, before disbursement.
Verify-before-trust	Server-to-server payment verification with exact amount/currency; webhooks are only triggers.
Idempotency	The property that retrying the same request has no additional effect.
Four-eyes	A second, different admin must approve a dispute refund.
NDPA	Nigeria Data Protection Act 2023.
FCCPC	Federal Competition and Consumer Protection Commission; regulates digital lending conduct.
CBN	Central Bank of Nigeria; licenses payment/escrow institutions.
PPR	Partial Prerendering (Next.js); static shell with streamed dynamic data.
FTS	PostgreSQL full-text search.
PgBouncer	Connection pooler used in front of Postgres at scale.
PITR	Point-in-time recovery for the database.

Prepared from the Goods.ng repository and verified build state on 8 November 2026. This document is a planning and acceptance baseline, not legal advice. Licensed partner terms, regulatory interpretations and go-live dates must be confirmed by counsel and the relevant providers.